Direct companion generators: complete finite validation

2026-11-01 · Exact component computation · direct-companion-validation

All 162 cases passed. The public generator constructor agrees with an independent dense feature construction across 27 coefficient-field contexts, three gamma labels, and two arms. The normalized dual-witness panel passed all 66 cases. All 1,038 corruption controls and 324 invalid-gamma rejections passed. Candidate and tuple-weight A/A outputs and counts agree exactly. The single producer invocation returned 0; no restart was used.

What was checked. The checker forms quadratic characters using square sets, then independently constructs the full weighted feature Gram matrix B and boundary matrix H. It checks B - gamma T B T^t = H and the companion identity Z_(r,gamma) A - A Z_(r,1) = G K^t entrywise. For orders at most 19 it also uses explicit companion matrices. A=D^{-1} J B J, where J reverses coordinates and D=diag(gamma^i). Coprimality follows from the nonzero difference of X^r-gamma and X^r-1; gamma^r=gamma is checked in every case.

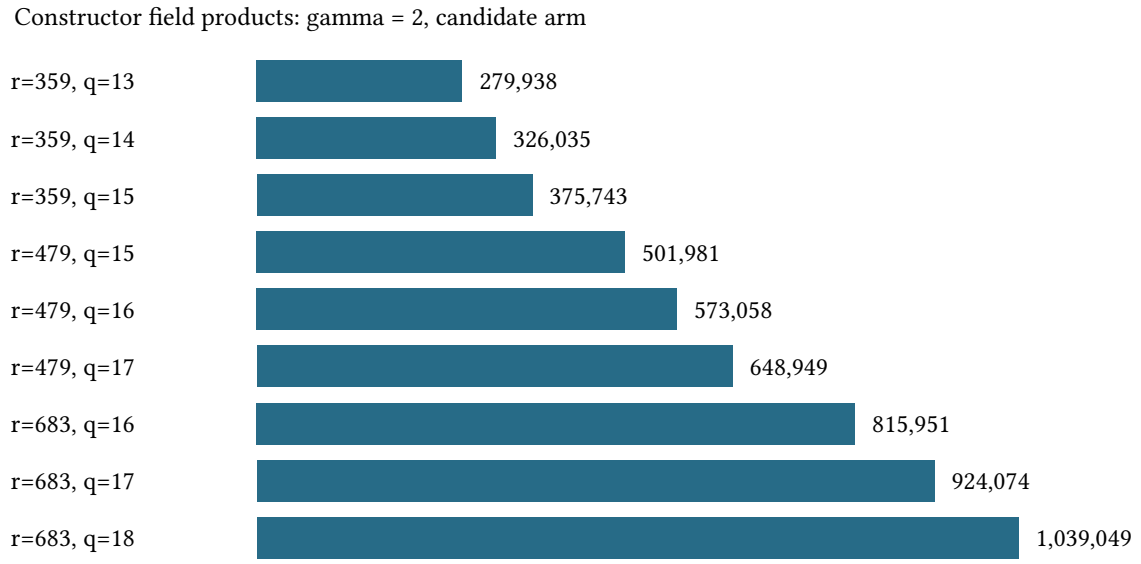
Transport. Every case checks a random known primal vector and a general residual-dual vector. For all width-one contexts and the retained public r13q4 and r479q15 exclusion witnesses, normalization and nullspace identities are checked on both sides. These are tests of transport, using known public witnesses; no structured solve backend is involved. Old-compiler labels, weights and inverse-Pascal boundary coordinates are checked independently.

Controls. Six deliberate changes per case alter G, K, the left wrap, inverse diagonal, RHS or field-product count. Each is rejected by full constructor-output/count equality against the independent reference. The 66 normalized cases also reject a zero witness. Gamma 0 and 1 are rejected in each case. These controls test the named validators, not every possible corruption of a future solver.

Panel. Original contexts: (7,6),(13,12),(19,15),(43,15),(73,15),(349,15). Larger contexts: r359 with q13–15; r479 with q15–17; r683 with q16–18. Width one is checked for all nine orders. Retained preflight contexts: (7,3),(13,4),(19,4). Gamma labels are 2, r-1, and a frozen seeded draw; coincident values retain their separate labels. Weights and random vectors use seed 20261101+100000r+100q+gamma-index. Gamma seed: 20261101+100r+q.

Custody and execution. Predata commit 8a4876f; nine source locks and the empty input lock are retained. Command: timeout 900 ../.audit-venv/bin/python -u direct-companion-validation/check.py. Exact row timestamps span 2026-11-01T19:48:17.341408+00:00 to 2026-11-01T19:49:34.381694+00:00. Shared local host; no valid timing, CPU or peak-RAM measurement was collected. All 162 planned cases completed; zero failed, interrupted or excluded cases. Protocol and export are checked separately.

Constructor accounting and next backend



Recorded field-product counts from runs/cases.jsonl, candidate instances 18,21,24,27,30,33,36,39,42. Data and SVG renderer are retained beside this PDF. Each bar is one exact case; uncertainty bars are not applicable. A/A covers all 81 matched instances, not just these nine illustrated rows.

Exact ledger. Let K be the number of offset patterns, N the feature count, s=2K+1=3+q(q-1), and pc(e)=popcount(e)+bitlength(e)-1. The checked constructor has r pc((r-1)/2) + 2r sum_P |P| + sum_P pc(q-max(P)) + N-1 + 2r + 2rs field products; 1+K+rs field additions; and one field inversion. The constructor retains O(rs) generator data. Dense-checker and old-reference work are excluded from this constructor ledger. No total compiler cost, runtime improvement or memory reduction is inferred from these counts.

Concrete next dependency. The companion representation and transport now have a complete finite certificate panel. The next component is a structured matrix multiplication and singular-system backend that consumes G,K and returns separately checkable primal solutions or normalized dual witnesses. The base-field preconditioning guarantee must be treated explicitly; the existing full prime-field recovery instrument remains a separate artifact.

Evidence paths. runs/summary.json; runs/cases.jsonl; runs/export.json; source-lock.json; protocol.json; producer.log; exit-code.txt; launch.json; export-check.log. The derivation and constructor are in ../direct-cyclic-companion/DERIVATION.md and generators.py. This report is a fresh result record and does not rewrite the earlier unexecuted-construction report. Provenance and file hashes are retained at the experiment root.